

Associations in Domain Modeling

Definition

An **association** is a relationship between classes (or their instances) indicating a meaningful connection. In UML, it is defined as the semantic relationship between classifiers involving instance connections.

When to Show an Association

Guideline

Include associations where knowledge of the relationship needs to be preserved for some duration (milliseconds to years). Examples:

- **Sale–SalesLineItem:** Needed to reconstruct a sale and calculate totals.
- **Ledger–Sale:** Records completed sales for legal/accounting purposes.
- **Monopoly:** Piece–Square, Player–Piece, Square–Board.

Avoid unnecessary associations (e.g., Cashier looking up ProductDescriptions does not need to be remembered).

UML Notation for Associations

UML Representation

- Represented as a line connecting classes.
- Optional **reading direction arrow** indicates how to read the association.
- Each end may have:
 - Role name
 - Multiplicity
 - Navigability
- Name format: **ClassName-VerbPhrase-ClassName**, starting with a capital letter.
- Examples:
 - **Sale Paid-by CashPayment** (better than “Sale Uses CashPayment”)
 - **Player Is-on Square** (better than “Player Has Square”)

Multiplicity

Multiplicity

- Defines how many instances of Class A can be associated with one instance of Class B.
- Example: **Store Stocks Item** → 1 store can stock many items (*).
- Multiplicity communicates domain constraints, which may be reflected in software.
- Example: “0..1” vs “1” depends on whether an Item must always belong to a Store instance.

Multiple Associations Between Classes

Multiple Associations

- Distinct associations may exist between the same classes.
- Example (Airline domain):
 - Flight → Airport: **Flies-to**
 - Flight → Airport: **Flies-from**

Common Associations List (Business Systems)

Common Associations

- Transaction related to another transaction: **CashPayment–Sale**
- Line item of a transaction: **SalesLineItem–Sale**
- Product/service for a transaction/line item: **Item–SalesLineItem, Flight–Reservation**
- Role related to transaction: **Customer–Payment, Passenger–Ticket**
- Part of: **Drawer–Register, Square–Board, Seat–Airplane**
- Contained in/on: **Register–Store, Item–Shelf, Passenger–Airplane**
- Description for: **ProductDescription–Item, FlightDescription–Flight**
- Known/logged/captured in: **Sale–Register, Piece–Square, Reservation–FlightManifest**
- Member of: **Cashier–Store, Player–MonopolyGame, Pilot–Airline**
- Organizational subunit: **Department–Store, Maintenance–Airline**

- Uses/manages/owns: **Cashier–Register, Player–Piece, Pilot–Airplane**
- Sequential/next to: **SalesLineItem–SalesLineItem, Square–Square, City–City**

Example: POS Domain Associations

Example

- Transactions related to another transaction: **Sale Paid-by CashPayment**
- Line items of a transaction: **Sale Contains SalesLineItem**
- Product for a transaction: **SalesLineItem Records-sale-of Item**